Lie Algebras and their Representations

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Lecture 1

1 Introduction

A *Lie group* is fundamentally a group, which also a (smooth) manifold. For example, $\mathrm{GL}_n, \mathrm{SL}_n, \mathrm{SO}_n, \mathrm{Sp}_{2n}$.

> **Example 1.1**
> A prototypical example of a Lie group is the circle group S^1.

Let G be a Lie group. Then the *Lie algebra* of G is the tangent space at the identity e of G. That is,

$$\mathfrak{g} = \mathrm{T}_e G$$

$\mathfrak{g}$ is a vector space, with additional structure, which we will see later.

By taking a derivative, we turn the conjugation map

$$\begin{aligned} G &\to \mathrm{Aut}(G) \\ g &\mapsto g(\cdot)g^{-1} \end{aligned}$$

into a map

$$\mathrm{ad} : \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$$

called the *adjoint*. This gives a bilinear map

$$\begin{aligned} [\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} &\to \mathfrak{g} \\ [x, y] &= \mathrm{ad}(x)(y) \end{aligned}$$

> **Example 1.2**
> If $G = \mathrm{GL}_n(\mathbb{R})$, then we have that $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{R}) = \mathrm{Mat}_n(\mathbb{R})$, with
>
> $$[x, y] = xy - yx$$

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What do Lie algebras tell us about the structure of the Lie group G?

- We will define the *root system* of $\mathfrak{g}$, and this then tells us about commutator relations in G (see Carter's book).
- We will define the *Weyl group* of $\mathfrak{g}$. For example, the Weyl group of $\mathfrak{gl}_n(\mathbb{C})$ is isomorphic to S_n, there is an embedding $S_n \hookrightarrow \mathrm{GL}_n(\mathbb{C})$, vie permutation matrices. Let B denote the Borel subgroup of upper triangular matrices in $\mathrm{GL}_n(\mathbb{C})$, then there exists a *Bruhat decomposition*

$$G = \bigsqcup_{w \in S_n} BwB$$

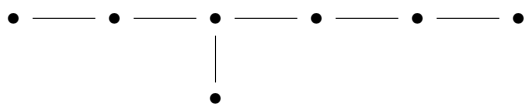
Lie algebras also give us information of the representation theory of G and $\mathfrak{g}$. For example, there exists a bijection

$$\{\text{finite dimensional } \mathbb{C}\text{-representations of } \mathrm{SL}_n(\mathbb{C})\} \leftrightarrow \{\text{finite dimensional } \mathbb{C}\text{-representations of } \mathfrak{sl}_n(\mathbb{C})\}$$

Moreover, we can describe the right hand side completely.

In addition, Lie algebras have applications in Algebraic Geometry, for example, we can use Lie algebras to build families of surfaces, of equivalenrly, algebraic curves (See book by Slodovy).

We will define the *Dynkin diagrams* of *semisimple Lie algebras*. For example,



is a Dynkin diagram of type E_7, and understanding the Dynkin diagram tells us about the singularities on the surfaces.

Moreover, Lie algebras also have applications in number theory, root systems/Weyl groups give the structure of groups over $\mathbb{Q}_p$, the p-adic integers (see paper on Moodle). Local Langlands correspondence predicts a relationship

$$\{\text{Galois theory of local fields}\} \leftrightarrow \{\text{complex Lie theory}\}$$

Finally, there any other applications, for example algebraic groups, quantum groups, theoretical physics, quantum mechanics.

2 Basic definitions and examples

Let k be a field. Most of the time, $k = \mathbb{C}$, but not always. We will sometimes point out how things can go wrong in characteristic p.

Definition 2.1 (Lie algebra)

A *Lie algebra* over k is a vector space $\mathfrak{g}$ over k, together with a bilinear pairing

$$[\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

satisfying

1. $[xx] = 0$ for all $x \in \mathfrak{g}$,
2. the Jacobi identity
$$[x[yz]] + [y[zx]] + [z[xy]] = 0$$

Notation 2.2. Note that when clear, we will write $[xy] := [x, y]$.

Remark 2.3. In particular, we have antisymmetry, i.e.

$$[xy] = -[yx]$$

Definition 2.4 ((Lie) subalgebra)

A k-vector subspace $\mathfrak{h}$ of $\mathfrak{g}$ is a *(Lie) subalgebra* if $\mathfrak{h}$ is closed under the Lie bracket of $\mathfrak{g}$. That is, for all $x, y \in \mathfrak{h}$, $[xy] \in \mathfrak{h}$.

Example 2.5

Let V be a finite dimensional k-vector space, then

1. Let $\mathfrak{gl}(V) = \mathrm{End}(V)$, with $[xy] = xy - yx$. If we choose a basis for V, then we can identify $\mathfrak{gl}(V)$ with $\mathrm{Mat}_n(k)$. In this case, we will write $\mathfrak{gl}_n$ or $\mathfrak{gl}_n(k)$.
2. Let
$$\mathfrak{sl}(V) = \{x \in \mathfrak{gl}(V) \mid \mathrm{tr}(x) = 0\}$$
This defines a subalgebra of $\mathfrak{gl}(V)$, called the *special linear Lie algebra*, with $\dim(\mathfrak{sl}(V)) = \dim(V)^2 - 1$. The standard basis of $\mathfrak{sl}(V)$ is given by
$$E_{i,j} \text{ for } i \neq j$$
$$E_{i,i} - E_{i+1,i+1}$$
We will often write $\mathfrak{sl}_n$ for this Lie algebra.

Lecture 2

Example 2.6 (continued) 3. Suppose $\mathrm{char}(k) \neq 2$, and suppose V is endowed with a symmetric (non-degenerate?) bilinear form
$$\langle \cdot, \cdot \rangle : V \times V \to k$$
Then define
$$\mathfrak{so}(V) = \{x \in \mathfrak{gl}(V) \mid \langle xv, w \rangle + \langle v, xw \rangle = 0 \text{ for all } v, w \in V\}$$
If $M \in \mathrm{GL}(V)$ is such that $\langle v, w \rangle = v^{\mathsf{T}} M w$, so
$$\mathfrak{so}(V) = \left\{x \mid Mx + x^{\mathsf{T}} M = 0\right\}$$
We usually take
$$M = \begin{cases} \begin{pmatrix} 0 & I_\ell \\ I_\ell & 0 \end{pmatrix} & n = 2\ell \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_\ell \\ 0 & I_\ell & 0 \end{pmatrix} & n = 2\ell + 1 \end{cases}$$
These are called the *orthogonal Lie algebra*, denoted $\mathfrak{so}_n$.

Remark 2.7. In the case $n = 2$, let

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

viewed as matrices (in $\mathfrak{sl}_2(\mathbb{C})$). Indeed, this is the standard basis of $\mathfrak{sl}_2(\mathbb{C})$. Note

$$[ef] = h$$
$$[he] = 2e$$
$$[hf] = -2f$$

We'll see that (in some sense) the structure of all semisimple Lie algebras comes from $\mathfrak{sl}_2(\mathbb{C})$.

4. Again suppose $\mathrm{char}(k) \neq 2$. Now suppose V is endowed with a non-degenerate skew-symmetric (or alternating) bilinear form $\langle \cdot, \cdot \rangle$, then

$$\mathfrak{sp}(V) = \{x \in \mathfrak{gl}(V) \mid \langle xv, w \rangle + \langle v, xw \rangle = 0\}$$

In coordinates, we take the form $\langle \cdot, \cdot \rangle$ to be the skew-symmetric form given by

$$M = \begin{pmatrix} 0 & I_\ell \\ -I_\ell & 0 \end{pmatrix}$$

where $n = 2\ell$. This is called the *symplectic Lie algebra*, denoted $\mathfrak{sp}_n$.

Exercise: Check that $\mathfrak{so}_n$ and $\mathfrak{sp}_n$ are Lie subalgebras of $\mathfrak{gl}_n$.

Example 2.8 (continued) 5. Any vector space V is a Lie algebra with $[vw] = 0$ for all v, w. We call such Lie algebras *abelian*. It is named like this, since for *linear Lie algebras*, that is, any subalgebra of $\mathfrak{gl}(V)$ where V is finite dimensional, $[xy] = xy - yx = 0$ is true if and only if x and y commute.

6. $\mathfrak{b}_n$ is the *Borel algebra* of upper triangular matrices

$$\begin{pmatrix} * & \cdots & * \\ & \ddots & \vdots \\ 0 & & * \end{pmatrix}$$

This is the Lie algebra associated to the Borel subgroup of upper triangular invertible matrices.

7. $\mathfrak{n}_n$ is the Lie algebra of strictly upper triangular matrices,

$$\begin{pmatrix} 0 & \cdots & * \\ & \ddots & \vdots \\ 0 & & 0 \end{pmatrix}$$

$\mathfrak{n}$ stands for *nilpotent*, see section 7.

3 Basic representation theory

Definition 3.1 (homomorphism, isomorphism)
A linear map $\varphi : \mathfrak{g} \to \mathfrak{h}$ between two Lie algebras is a *homomorphism* if

$$\varphi([xy]) = [\varphi(x), \varphi(y)]$$

for all $x, y \in \mathfrak{g}$. If φ is a linear isomorphism, we call φ an *isomorphism of Lie algebras*.

Definition 3.2 (representation)
A *representation* of $\mathfrak{g}$ is a Lie algebra homomorphism

$$\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$$

for some vector space V.

Notation 3.3. We also call V itself a *representation*, or a $\mathfrak{g}$-module. We write $\mathfrak{g} \curvearrowright V$, and say $\mathfrak{g}$ acts on V. We will write

$$x \cdot v = xv := \varphi(x)(v)$$

The *dimension* of the representation is the dimension of V.

Example 3.4 1. Let $\dim(V) = 1$, then $\mathfrak{g} \curvearrowright V$ via $xv = 0$. This is called the *trivial representation*.

2. $\mathfrak{g}$ is a subalgebra of $\mathfrak{gl}(V)$, then the natural inclusion $\mathfrak{g} \hookrightarrow \mathfrak{gl}(V)$ is called the *defining representation*.

3. Let $x \in \mathfrak{g}$, define $\mathrm{ad}_x : \mathfrak{g} \to \mathfrak{g}$ by $\mathrm{ad}_x(y) = [xy]$. The map

$$\begin{aligned} \mathrm{ad} : \mathfrak{g} &\to \mathfrak{gl}(\mathfrak{g}) \\ x &\mapsto \mathrm{ad}_x \end{aligned}$$

is called the *adjoint representation*.

Remark 3.5. Recall e, h, f from remark 2.7, the adjoint representation of $\mathfrak{sl}_2(\mathbb{C})$ has matrices

$$\mathrm{ad}_h = \begin{pmatrix} 2 & & \\ & 0 & \\ & & -2 \end{pmatrix} \quad \mathrm{ad}_e = \begin{pmatrix} & -2 & \\ & & 1 \\ & & \end{pmatrix} \quad \mathrm{ad}_f = \begin{pmatrix} -1 & & \\ & 2 & \\ & & \end{pmatrix}$$

with respect to the basis $\{e, h, f\}$.

4. If V, W are representations of $\mathfrak{g}$, then so is their *direct sum* $V \oplus V$, via

$$x(v, w) = (xv, xw)$$

5. If V is a representation of $\mathfrak{g}$, then so is the *dual* V^*,

$$(xf)(v) = -f(xv)$$

for all $x \in \mathfrak{g}, f \in V^*, v \in V$.

6. If V, W are representations of $\mathfrak{g}$, then so is the *homomorphisms* $\mathrm{Hom}(V, W)$, via

$$(x \cdot f)(v) = x \cdot f(v) - f(x \cdot v)$$

Definition 3.6 (equivariant, isomorphism)
If V, W are representations of $\mathfrak{g}$, then a linear map $\varphi : V \to W$ is called $\mathfrak{g}$-*equivariant* for all $x \in \mathfrak{g}, v \in V$,

$$x \cdot \varphi(v) = \varphi(x \cdot v)$$

We say V, W are *isomorphic* if there exists a $\mathfrak{g}$ equivariant isomorphism (of vector spaces) $V \to W$.

Definition 3.7 (subrepresentation, irreducible)
A *subrepresentation* $W \leq V$ is a subspace $xw \in W$ for all $x \in \mathfrak{g}, w \in W$. A non-zero representation V

is *irreducible* or *simple* if the only subrepresentations of V are 0 and V.

Exercise: The trivial representation is irreducible, and so are the defining and the adjoint representation of $\mathfrak{sl}_2(\mathbb{C})$.

Definition 3.8 (completely reducible, semisimple)
A representation V is *completely reducible*, or *semisimple* if it decomposes as the direct sum of irreducible representations.

Exercise: A representation V is completely reducible if and only if for every subrepresentation W of V, there exists another subrepresentation W' such that

$$V = W \oplus W'$$

Example 3.9
If V is a representation, $W \leq V$ a subrepresentation, then the *quotient* V/W is a representation of $\mathfrak{g}$, via

$$x(v + W) = (xv) + W$$

Remark 3.10. The isomorphism theorems hold for quotient representations as well.

4 Representation theory of $\mathfrak{sl}_2(\mathbb{C})$

Recall

$$\mathfrak{sl}_2(\mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \;\middle|\; a, b, c \in \mathbb{C} \right\}$$

Example 4.1
We have

$$\mathfrak{b}_2 = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right\}$$

which is the Borel subalgebra of 2×2 matrices. Let $V = \text{span}\left\{ v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$, and the defining representation of $\mathfrak{b}_2$. This is not completely reducible. To see this, set

$$V_1 = \text{span}\{v_1\}$$

and this does not have a complement.

Definition 4.2 (faithful)
A representation V of $\mathfrak{g}$ is *faithful* if the map

$$\mathfrak{g} \to \mathfrak{gl}(V)$$

is injective.

From now on, all Lie algebras and their representations will be over C, unless stated otherwise.
Let V be a representation of $\mathfrak{sl}_2(\mathbb{C})$. Recall the basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

of $\mathfrak{sl}_2(\mathbb{C})$. We know three representations of $\mathfrak{sl}_2(\mathbb{C})$ already.

dimension	name	action of h
1	trivial	0
2	defining	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
3	adjoint	$\begin{pmatrix} 2 & & \\ & 0 & \\ & & -2 \end{pmatrix}$

Definition 4.3 (weight space)
For $\lambda \in \mathbb{C}$, the *λ-weight space* of V is

$$V_\lambda = \{v \in V \mid hv = \lambda v\}$$

is the λ-eigenspace of h.

The following are vector space sums, not decompositions into subrepresentations.

- For the trivial representation,
$$V = V_0$$
- For the defining representation,
$$V = V_1 \oplus V_{-1}$$
- For the adjoint representation,
$$V = V_2 \oplus V_0 \oplus V_{-2}$$
where $V_2 = \langle e \rangle$, $V_0 = \langle h \rangle$, $V_{-2} = \langle f \rangle$.

For $v \in V_\lambda$,

$$h(ev) = (he)v = ([he] + eh)v = 2ev + \lambda ev = (\lambda + 2)ev$$

Hence $ev \in V_{\lambda+2}$. Similarly, $fv \in V_{\lambda-2}$.

That is, we have

$$\cdots \underset{f}{\overset{e}{\rightleftarrows}} V_{\lambda-2} \underset{f}{\overset{e}{\rightleftarrows}} V_\lambda \underset{f}{\overset{e}{\rightleftarrows}} V_{\lambda+2} \underset{f}{\overset{e}{\rightleftarrows}} \cdots$$

Definition 4.4 (highest weight vector)
A non-zero

$$v \in V_\lambda \cap \ker(e)$$

for some λ is called a *highest weight vector (of weight λ)*

Example 4.5
In the adjoint representation, e is a highest weight vector.

Lemma 4.6. Suppose $v \in V_\lambda$ is a highest weight vector. Then for all $n \geq 1$,

$$ef^n v = n(\lambda - n + 1)f^{n-1}v$$

Proof. Induction on n. For $n = 1$,

$$(ef)v = ([ef] + fe)v = (h + fe)v = \lambda v = 1(\lambda - 1 + 1)f^0 v$$

The inductive step follows similarly. □

Lemma 4.7. Suppose $v \in V_\lambda$ is a highest weight vector. Then

$$W = \text{span}\{v, fv, f^2v, \dots\}$$

is a sub-representation of V,

Proof. Suffices to show that for $w = f^n v \in W$, then $ew, fw, hw \in W$. By definition, $fw \in W$ is obvious. $ew \in W$ follows by lemma 4.6, and the $n = 0$ case is just $ew = 0$.

For hw,

$$f^n v \in V_{\lambda - 2n}$$

and so

$$hw = (\lambda - 2n)w \in W$$

□

Proposition 4.8. If V is finite dimensional, then a highest weight vector exists.

Proof. Choose any nonzero eigenvector v of h (always exists as we are working over $\mathbb{C}$). Consider

$$v, ev, e^2v, \dots$$

These are eigenvectors for h, with distinct eigenvalues. Hence the set

$$\{e^n v \mid e^n v \neq 0\}$$

is linearly independent. As V is finite dimensional, this set must be finite. Hence there must exists n such that

$$e^n v \neq 0 \quad \text{and} \quad e^{n+1} v = 0$$

Then $e^n v$ is a highest weight vector. □

Lemma 4.9. Suppose V is finite dimensional, and $v \in V_\lambda$ is a highest weight vector, then $\lambda \in \mathbb{Z}_{\geq 0}$.

Proof. Any non-zero vectors of the form $f^n v$ must be linearly independent, so there exists $n \geq 0$ such that $f^n v \neq 0, f^{n+1} v = 0$. By lemma 4.6,

$$0 = ef^{n+1} v = (n+1)(\lambda - n) f^n v$$

Hence we must have that $\lambda = n$, since $n + 1 \neq 0, f^n v \neq 0$. □

Conclusion: Suppose V is irreducible, of dimension $n + 1$. Then by proposition 4.8, a highest weight vector $v \in V_\lambda$ exists. By lemma 4.7, we have a subrepresentation

$$W = \text{span}\{v, fv, \dots\}$$

So by irreducibility,

$$\{v, fv, \dots, f^n v\}$$

is a basis, as the $f^i v$ are linearly independent, and we know from lemma 4.9 that $\lambda = n$.

Corollary 4.10. If V is an irreducible representation of $\mathfrak{sl}_2$, of dimension $n + 1$, then there exists a basis $v_0, \dots, v_n$ of V, such that the actions are:

$$hv_i = (n - 2i)v_i \qquad fv_i = \begin{cases} v_{i+1} & i + 1 \leq n \\ 0 & i = n \end{cases} \qquad ev_i = \begin{cases} i(n - i + 1)v_{i-1} & i - 1 \geq 0 \\ 0 & i = 0 \end{cases}$$

In particular, there is a unique irreducible representation of $\mathfrak{sl}_2$ with dimension $n + 1$ for all $n \geq 0$.

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